

In early AI systems, like AM and Eurisko, actual progress came only through making modifications to small segments of the LISP code that represented mathematical ideas, and it was not just luck; it was that the structure of LISP happened to align with the structure of the way the concepts of mathematics are organized. Because of this alignment, the programmer was able to make small modifications to the code and discover new and interesting ideas, such as modifying the specific predicate LIST-EQUAL to the general predicate SAME-LENGTH. This is what representation is all about: when the representation structure of the data aligns with the conceptual structure of its meaning, then it becomes easier to search for discoveries. In today's terminology, this relates to model selection; in any field, a researcher or engineer needs to pick a model or some language, which at least gives the opportunity to explore a meaningful space.

But the paper also points to an important problem: just because something is syntactically correct does not mean it has meaning. In trying to evolve heuristics rather than math, when AM applied the mutational strategy it used for math to produce heuristics, things fell apart. With heuristics, the programs were longer and more complex, and did not retain the structure of the real-world meaning behind heuristics. Therefore, most of what it produced as "new" had no meaning. This is a clear case of type III error - getting a correct technical answer to the wrong question. The system was still operating and producing output, but it was just not solving problems worth solving.

What are we to take away from this? In short, meaningful discovery is based on an appropriate correspondence between form (syntax) and content (semantics). AM worked in part because the two corresponded well in mathematics, but when the correspondence was lost, the discoveries

made through AM became meaningless. It follows that Type III error introduces itself when we mistake formally correct discoveries for meaningful discoveries. To guard against this error, we must design our data and models carefully to maintain a grounding system of meaningful goals.